Scalability & Future Plan

Date	29 September 2026
Team ID	SWTID-2026-8642
Project Name	Import Data Using Transform Maps (Spreadsheet)
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Manual file upload requirement for spreadsheet imports.	Causes delays in data updates and increases the risk of human error.	High
2	Performance degradation in transform maps during large dataset imports.	Increases system processing time and impacts instance performance.	Medium
3	Limited data validation rules configured in the staging table.	Invalid rows get loaded into the staging table, causing downstream errors later.	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Manual spreadsheet uploads via admin interface.	Set up Scheduled Data Sources and Automated Email/FTP Integration.
2	Data Storage	Accumulation of temporary records in the staging table.	Use Import Set Table Cleanup Scheduled Jobs to periodically clear staging logs.
3	Performance	Single-threaded sequential record transformation.	Enable Concurrent Import Sets and Parallel Processing to increase execution speed.

4	Security	Basic role-based access for import set tables.	Implement Advanced ACLs and REST API authentication flows.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Automated FTP/SFTP Integration for daily spreadsheet sync.	Q2 2026	Completely eliminates manual spreadsheet upload tasks.
Phase 3	Integration Hub & REST API based real-time data flow.	Q3 2026	Enables real-time record synchronization without file dependency.
Phase 4	Advanced Scripting & Automated Error Alerting Notifications.	Q4 2026	Instantly notifies data transform failures and reduces resolution time.